

Who goes bankrupt next?

Three tuned classifiers screen 6,819 Taiwanese companies for bankruptcy risk, so a bank analyst knows which loan applicants to check first.

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AI for Good · Hackathon 5: Model Showdown · KNN, logistic regression, random forest

1 · PROBLEM DEFINITION

Bankruptcy is rare, so it is easy to miss

WHAT Some companies applying for a business loan are already heading for bankruptcy, and it is not obvious from their accounts.

FOR WHOM Bank risk analysts, who must decide which applicants need a deeper check.

WHERE Companies listed in Taiwan, the population of our data (1999–2009).

WHEN Before the lending decision, while the risk can still be avoided.

+6%

global business insolvencies
expected in 2025

+5%

a further rise forecast for 2026;
Asia is half of the global increase

3.2%

of companies in our data went
bankrupt: 220 of 6,819

96.8%

accuracy from always guessing
"not bankrupt"

Safer lending protects jobs and income

8

DECENT WORK
AND ECONOMIC
GROWTH

The specific outcome. Whether a listed Taiwanese company will be classified as bankrupt, predicted from its financial ratios, so a bank can review risky business-loan applicants before it lends.

Target 8.10 "Strengthen the capacity of domestic financial institutions to encourage and expand access to banking, insurance and financial services for all."

Who benefits

Banks lend with fewer surprises and keep credit available.
Employees depend on financially stable employers for their jobs and income.

What it does not do

It does not predict job loss for individuals, prevent a bankruptcy or decide a loan. It gives an analyst an earlier signal.

Bank risk analysts assessing business loans

SITUATION	They already read company financials and judge the risk of every business-loan application.
NEED	To know which applicants need a deeper investigation first, because time per file is limited.
USE	The model flags likely bankruptcies and gives a probability. The analyst reviews, then decides.
AFFECTED	The companies being scored, and the employees whose jobs depend on them.
MISSING	Companies outside Taiwan, anything after 2009, and any data on the workers themselves.

NOT FOR

- Automatic loan decisions**
A flag is a reason to look closer, never a verdict.
- Companies outside the data**
Other countries or today's market need new validation.
- Personal investors**
It was not built or tested for buying shares.
- Replacing the analyst**
Professional judgement stays in charge.

DATASET CARD

6,819 real companies, 95 financial ratios

SOURCE	UCI Machine Learning Repository, Taiwanese Bankruptcy Prediction (DOI 10.24432/C5004D)
COLLECTED	From the Taiwan Economic Journal, financial years 1999–2009
TARGET	Bankrupt? (1 = bankrupt), defined by the business rules of the Taiwan Stock Exchange
FEATURES	95 ratios: profitability, liquidity, leverage, cash flow, growth, asset efficiency
QUALITY	No missing values, no duplicate rows
LICENCE	CC BY 4.0, so the 11 MB file is in our repo

Class balance

Not bankrupt: 6,599 (96.8%)



Bankrupt: 220 (3.2%)



Known limitations

One country, listed companies only, 1999–2009.
Only financial ratios: no management, market or qualitative information.

3 · SOLUTION DESCRIPTION

One notebook, from raw file to prediction

1

Split first
80/20, stratified,
random_state 42

2

Baseline
Dummy that always
says "not bankrupt"

3

Pipeline
Scaler fitted inside
each fold

4

Tune
GridSearchCV, 5
folds, recall

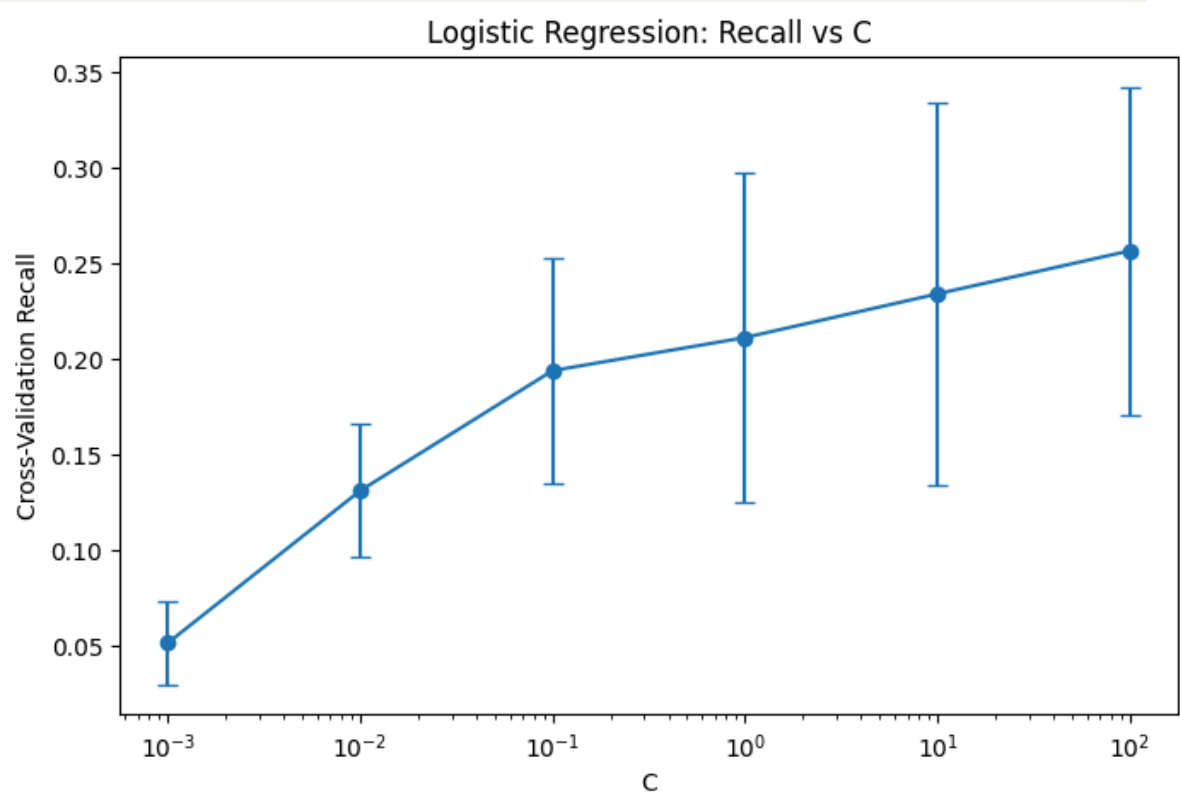
5

Test once
1,364 unseen
companies, 44
bankrupt

6

Predict
Class and
probability for a
new company

Model	Hyperparameters tuned	Why this setup
KNN	n_neighbors: 3, 5, 7, 9, 11, 15, 21	Distances need scaling
Logistic regression	C: 0.001, 0.01, 0.1, 1, 10, 100	Penalty needs scaling
Random forest	trees 100/200 · max_depth None/10/20 · min_samples_split 2/5	class_weight balanced for the rare class



Same folds, same metric, training data only for all three models · the test set is touched once, at the end

MODEL COMPARISON

The random forest finds the most bankruptcies

Model	Best settings	Train recall	CV recall (± spread)	Test precision	Test recall	Test F1
Baseline (dummy)	always "not bankrupt"	0.000	–	0.000	0.000	0.000
KNN	n_neighbors 3	0.375	0.194 ± 0.062	0.421	0.182	0.254
Logistic regression	C 100	0.307	0.257 ± 0.086	0.333	0.159	0.215
Random forest (recommended)	200 trees, max_depth 10, min_samples_split 5	1.000	0.449 ± [__]	0.410	0.568	0.476

The lead is real

CV recall 0.449 against 0.257: a gap of 0.19, larger than the spread over the folds.

But it overfits

Train recall 1.000 against 0.449 in CV: it memorises the training companies.

The others fall short

Low recall on training, CV and test data: both miss more than 4 in 5 test bankruptcies.

CONFUSION MATRIX • RANDOM FOREST

25 of 44 bankruptcies caught, 19 missed

	Predicted: not bankrupt	Predicted: bankrupt
Actually not bankrupt	1,284 correctly passed	36 false alarms: extra review
Actually bankrupt	19 missed: looked safe	25 caught: flagged in time

How to read it: rows = what really happened · columns = what the model predicted

Test set: 1,364 companies the model never saw during training or tuning, 44 of them bankrupt

Recall 0.57

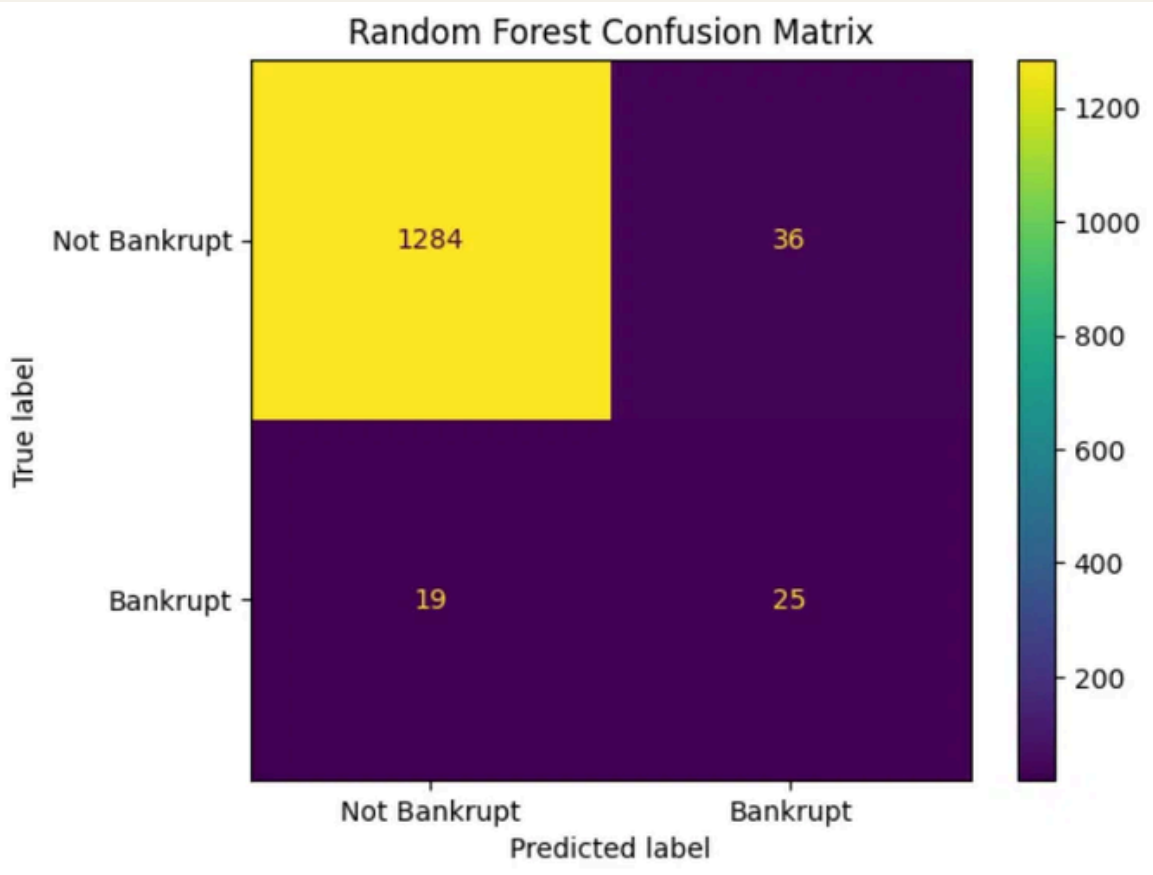
More than half of the failing companies are flagged before a loan decision.

Precision 0.41

61 companies flagged, 25 really failed: a review list an analyst can work through.

The 19 misses

Their ratios looked healthy. This is the error that matters most, so the analyst keeps the final say.



A missed bankruptcy costs more than a false alarm

Missed bankruptcy (false negative)

The bank lends to a company that fails: the loan is lost and the company's workers lose their jobs anyway.

False alarm (false positive)

A healthy company gets an extra review: analyst time, and a slower answer for the applicant.

So recall is our main metric: we choose the model that misses the fewest bankruptcies, and check precision so the list stays workable.

Test recall: share of 44 bankruptcies found

Baseline: 0.00 (finds 0)



Logistic regression: 0.16 (finds 7)



KNN: 0.18 (finds 8)



Random forest: 0.57 (finds 25)



Why machine learning: the simple alternative finds nothing, and no analyst can weigh 95 ratios at once by hand.

THE DECISION WE WOULD DEFEND HARDEST

Recall, not accuracy.

Dummy baseline

96.8% accurate

catches 0 of 44 bankruptcies

Random forest

96.0% accurate

catches 25 of 44 bankruptcies

One new company, one prediction

```
new_company = X_test.iloc[[0]]  
model = rf_grid.best_estimator_  
model.predict(new_company)  
model.predict_proba(new_company)
```

The input is all 95 financial ratios of one company the model never saw during training or tuning. The output is a class and a probability for each class.

Restart and Run All takes the raw data.csv to the comparison table and this prediction, with no manual steps.

FINAL MODEL PREDICTION

Lower bankruptcy risk

Probability of bankruptcy

0.96%

Probability of not bankrupt

99.04%

For the analyst: no extra review needed, final call is theirs.

A wrong flag affects a company and its workers

Risk	Consequence	What we did	Next step
Missed: 19 of 44 bankruptcies	Bank lends to a failing company; jobs lost	Recall as main metric; balanced weights in the forest	Lower the threshold to catch more
False alarm: 36 healthy firms	A healthy company's loan is delayed	Decision support only: the analyst decides	Show which ratios caused a flag
Narrow data: Taiwan, 1999–2009	Wrong for other countries or today	A "Not for" list in our README	Retrain and validate on recent data
No group data	Can't check if errors hit some sectors harder	Stated openly as a limitation	Add sector and size to check per group

25 settings, 125 cross-validation fits, one choice

1

Pick data

UCI, licence checked

2

Explore

3.2% bankrupt

3

Split

then baseline

4

Tune

3 models, recall

5

Evaluate

test set once

What went wrong, and the fix

Data path. The notebook loaded data.csv from a Colab-only folder and failed elsewhere. It now reads the file straight from our GitHub repo.

Overfitting. The random forest scores perfect recall on its training data, so we judged it only on cross-validation and the test set.

Class weights. Logistic regression had none and found only 7 of 44. Next time it gets class_weight balanced too.

25

hyperparameter settings: 7 KNN, 6 logistic regression, 12 random forest

125

fits: every setting on the same 5 folds

OUR RECOMMENDATION: THE RANDOM FOREST, AS A SCREENING TOOL

Not a verdict. A watch list.

Flag early · Judge on recall · The analyst decides

Questions?

Notebook, data, dataset card and README: github.com/pavitrapradeep/ai4g-portfolio-pavitra › Term 1 › Week 5 › hackathon